

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Artificial Intelligence and Data Science
ASSESSMENT TOOL 1 – Data Interpretation

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO4 – Precision	Assessment:	Assessment Tool 1 – Data Interpretation
Weightage:	35% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	B.Tech(AI&DS)	Date:	19/08/2026

Code of Conduct

I **VIMALA K 192424276** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:



Instructions

- Read each data set, table, semantic representation, or scenario carefully before answering.
- Analyze the given information and provide logical, evidence-based interpretations.
- Show all intermediate reasoning, semantic analysis steps, predicate representations, or calculations wherever applicable.
- Answers should be concise, structured, and technically accurate.
- Support your conclusions using the data provided in the question.
- Draw diagrams, semantic networks, parse trees, or logical representations wherever relevant.
- Avoid copying definitions alone; focus on analyzing the given scenario and interpreting the data.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic representations, identify action–entity relationships, detect interpretation errors, and evaluate meaning representation in chatbot systems.	Q1
First-Order Predicate Calculus & Representing Linguistically Relevant Concepts	Construct predicate representations, apply logical inference rules, derive conclusions, and evaluate reasoning in smart manufacturing environments.	Q2
Lexemes and Their Senses & Word Sense Disambiguation	Analyze contextual clues, determine correct word senses, evaluate ambiguity resolution, and improve semantic understanding in search systems.	Q3
Syntax-Driven Semantic Analysis	Analyze syntactic structures, assign semantic roles, evaluate parsing accuracy, and improve semantic interpretation in healthcare NLP applications.	Q4

Annexure A – Data Interpretation

1. Frame Semantics in Banking Customer-Service Systems

A digital banking platform uses semantic frames to understand customer requests. The NLP system identifies the **action, agent, object, and destination** involved in each request.

Customer Query	Generated Semantic Frame
"Transfer ₹20,000 from my savings account to my son's account."	TRANSFER(SourceAccount, Amount, DestinationAccount, Customer)
"Block my debit card immediately."	BLOCK(DebitCard, Customer)
"Send my transaction statement to my email."	SEND(TransactionStatement, Email, Customer)
"Increase my daily withdrawal limit."	MODIFY(WithdrawalLimit, Customer)

Additional transaction logs show:

Query ID	Actual User Intent	System Interpretation
B1	Transfer money to another account	Transfer money
B2	Block debit card	Block debit card
B3	Receive transaction statement by email	Send statement to email
B4	Increase withdrawal limit	Decrease withdrawal limit

Tasks:

1. Analyze the semantic frame of each banking request and identify the relationship between the action and its participants.
2. Identify the query in which the semantic interpretation is incorrect and explain why.
3. Analyze how incorrect identification of semantic roles can lead to an incorrect banking operation.
4. Recommend a frame-based semantic analysis approach to improve the reliability of banking conversational systems.

```
print("=" * 60)
print("FRAME SEMANTICS IN BANKING")
print("=" * 60)
```

```
requests = [
    ("Transfer Rs.20,000 from my savings account to my son's account.",
     "TRANSFER(SourceAccount, Amount, DestinationAccount, Customer)"),

    ("Block my debit card immediately.",
     "BLOCK(DebitCard, Customer)"),

    ("Send my transaction statement to my email.",
     "SEND(TransactionStatement, Email, Customer)"),

    ("Increase my daily withdrawal limit.",
     "MODIFY(WithdrawalLimit, Customer)")
]
```

```
for query, frame in requests:
    print("\nCustomer Query:")
```

```
print(query)
    print("Semantic Frame:")
    print(frame)
```

```
print("\nTASK 1")
print("""
Transfer:
Action = Transfer
Source = Savings Account
Amount = Rs.20,000
Destination = Son's Account
Customer = Account Holder
```

```
Block:
Action = Block
Object = Debit Card
Customer = Account Holder
```

```
Send:
Action = Send
Object = Transaction Statement
Destination = Email
Customer = Account Holder
```

```
Modify:
Action = Increase
Object = Withdrawal Limit
Customer = Account Holder
""")
```

```
print("TASK 2")
print("B4 is incorrect.")
print("Actual Intent: Increase withdrawal limit")
print("System Interpretation: Decrease withdrawal limit")
print("Reason: Increase and decrease have opposite meanings.")
```

```
print("\nTASK 3")
print("""
Incorrect semantic-role identification can cause the banking
system to perform the wrong operation.
```

```
For example, if increase withdrawal limit is interpreted as
decrease withdrawal limit, the customer's limit may be reduced.
""")
```

```
print("TASK 4")
print("""
A reliable banking NLP system should use:
1. Intent Classification
2. Named Entity Recognition
```

- 3. Semantic Role Labeling
 - 4. Frame-Based Semantic Parsing
 - 5. Business Rule Validation
 - 6. Transaction Validation
 - 7. Confirmation for important transactions
- ```
""")print("=" * 60)
print("BANKING ANALYSIS COMPLETED")
print("=" * 60)
```

```
co4 at1 q1.py - C:/Users/ADMIN/OneDrive/ドキュメント/NLP/ass python/co4 at1 q1.py (3.14.3)
File Edit Format Run Options Window Help

print("=" * 60)
print("FRAME SEMANTICS IN BANKING")
print("=" * 60)

requests = [
 ("Transfer Rs.20,000 from my savings account to my son's account.",
 "TRANSFER(SourceAccount, Amount, DestinationAccount, Customer)"),
 ("Block my debit card immediately.",
 "BLOCK(DebitCard, Customer)"),
 ("Send my transaction statement to my email.",
 "SEND(TransactionStatement, Email, Customer)"),
 ("Increase my daily withdrawal limit.",
 "MODIFY(WithdrawalLimit, Customer)"),
]

for query, frame in requests:
 print("\nCustomer Query:")
 print(query)
 print("Semantic Frame:")
 print(frame)

print("\nTASK 1")
print("""
Transfer:
Action = Transfer
Source = Savings Account
Amount = Rs.20,000
Destination = Son's Account
Customer = Account Holder""")

IDLE Shell 3.14.3
File Edit Shell Debug Options Window Help
Python 3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)] c
win32
Enter "help" below or click "Help" above for more information.

===== RESTART: C:/Users/ADMIN/OneDrive/ドキュメント/NLP/ass python/co4 at1 q1.p
=====

FRAME SEMANTICS IN BANKING

Customer Query:
Transfer Rs.20,000 from my savings account to my son's account.
Semantic Frame:
TRANSFER(SourceAccount, Amount, DestinationAccount, Customer)

Customer Query:
Block my debit card immediately.
Semantic Frame:
BLOCK(DebitCard, Customer)

Customer Query:
Send my transaction statement to my email.
Semantic Frame:
SEND(TransactionStatement, Email, Customer)

Customer Query:
Increase my daily withdrawal limit.
Semantic Frame:
MODIFY(WithdrawalLimit, Customer)
```

2. First-Order Predicate Calculus for Smart Agriculture

A smart agricultural system monitors crop fields using sensors and logical rules.

Field Data:

| Field | Soil Condition | Irrigation Status | Crop  |
|-------|----------------|-------------------|-------|
| F1    | Dry            | Available         | Rice  |
| F2    | Wet            | Available         | Wheat |
| F3    | Dry            | Unavailable       | Maize |
| F4    | Wet            | Unavailable       | Rice  |

The system uses the following predicate rules:

- $Dry(x) \wedge IrrigationAvailable(x) \rightarrow NeedsWater(x)$
- $NeedsWater(x) \rightarrow Irrigate(x)$
- $Wet(x) \rightarrow \neg NeedsWater(x)$
- $IrrigationUnavailable(x) \rightarrow \neg Irrigate(x)$
- $Rice(x) \wedge NeedsWater(x) \rightarrow PriorityIrrigation(x)$

Tasks:

1. Represent the field observations using First-Order Predicate Calculus.
2. Using the given rules, determine which fields require irrigation.
3. Determine whether F1 and F3 should be treated differently by the automated irrigation controller.
4. Analyze whether predicate logic alone is sufficient for agricultural decision support when sensor information is incomplete or uncertain.

```

print("=" * 60)
print("FIRST-ORDER PREDICATE CALCULUS")
print("SMART AGRICULTURE")
print("=" * 60)

fields = {
 "F1": {"soil": "Dry", "irrigation": "Available", "crop": "Rice"},
 "F2": {"soil": "Wet", "irrigation": "Available", "crop": "Wheat"},
 "F3": {"soil": "Dry", "irrigation": "Unavailable", "crop": "Maize"},
 "F4": {"soil": "Wet", "irrigation": "Unavailable", "crop": "Rice"}
}

print("\nTASK 1")

print("""
Dry(F1)
IrrigationAvailable(F1)
Rice(F1)

Wet(F2)
IrrigationAvailable(F2)
Wheat(F2)

Dry(F3)
IrrigationUnavailable(F3)
Maize(F3)

Wet(F4)
IrrigationUnavailable(F4)
Rice(F4)
""")

print("PREDICATE RULES")
print("1. Dry(x) AND IrrigationAvailable(x) -> NeedsWater(x)")
print("2. NeedsWater(x) -> Irrigate(x)")
print("3. Wet(x) -> NOT NeedsWater(x)")
print("4. IrrigationUnavailable(x) -> NOT Irrigate(x)")
print("5. Rice(x) AND NeedsWater(x) -> PriorityIrrigation(x)")

print("\nTASK 2")

if fields["F1"]["soil"] == "Dry" and fields["F1"]["irrigation"] == "Available":
 print("F1 -> Needs Water -> Irrigate")

```

```

else:
 print("F1 -> Do Not Irrigate")

if fields["F2"]["soil"] == "Dry" and fields["F2"]["irrigation"] == "Available":
 print("F2 -> Needs Water -> Irrigate")
else:
 print("F2 -> Do Not Irrigate")

if fields["F3"]["soil"] == "Dry" and fields["F3"]["irrigation"] == "Available":
 print("F3 -> Needs Water -> Irrigate")
else:
 print("F3 -> Cannot Irrigate")

if fields["F4"]["soil"] == "Dry" and fields["F4"]["irrigation"] == "Available":
 print("F4 -> Needs Water -> Irrigate")
else:
 print("F4 -> Do Not Irrigate")

print("\nTASK 3")
print("""
F1 and F3 are both dry.

F1 has irrigation available, so it can be irrigated.

F3 has irrigation unavailable, so it cannot be irrigated.

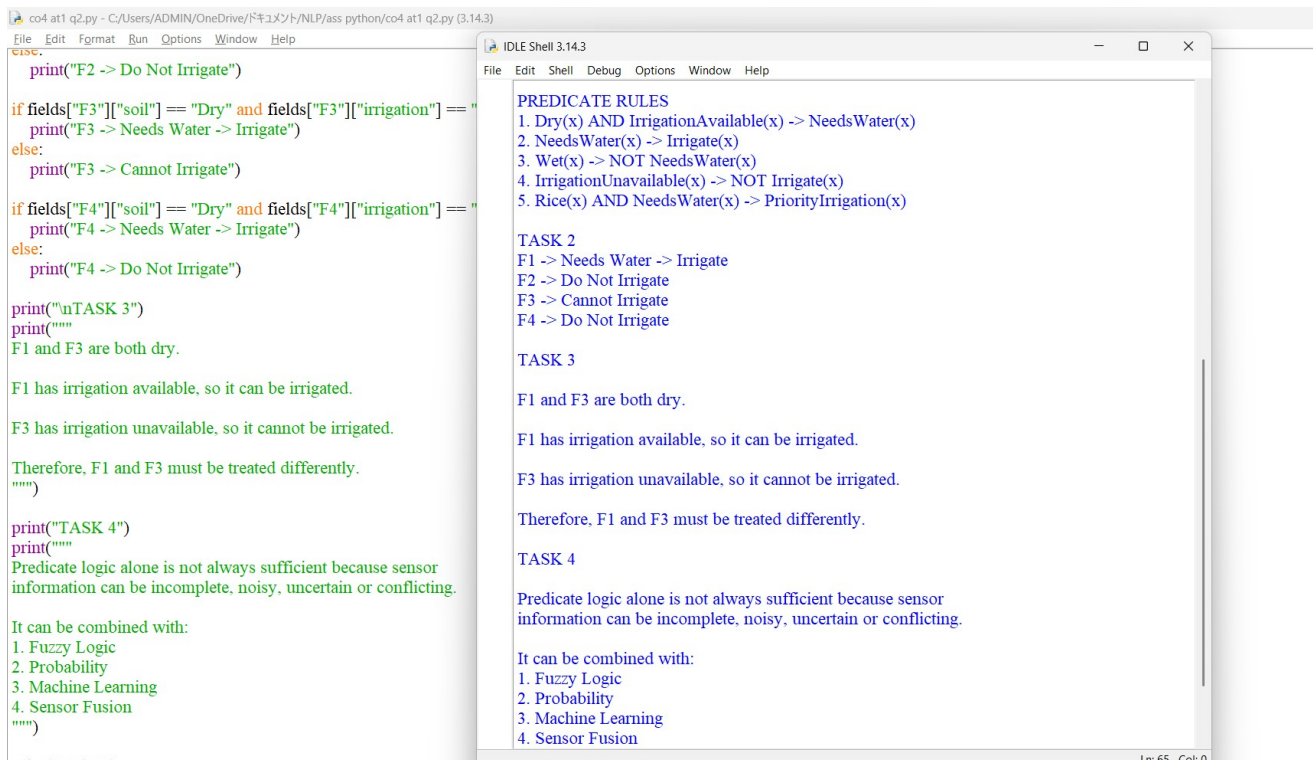
Therefore, F1 and F3 must be treated differently.
""")

print("TASK 4")
print("""
Predicate logic alone is not always sufficient because sensor
information can be incomplete, noisy, uncertain or conflicting.

It can be combined with:
1. Fuzzy Logic
2. Probability
3. Machine Learning
4. Sensor Fusion
""")

print("=" * 60)
print("AGRICULTURE ANALYSIS COMPLETED")

```



### 3. Word Sense Disambiguation in a Travel Recommendation System

A travel platform receives search queries containing ambiguous words. The recommendation engine uses surrounding words and user interactions to determine the intended meaning.

| Search Query     | Possible Sense 1  | Possible Sense 2          |
|------------------|-------------------|---------------------------|
| "Amazon tour"    | Amazon rainforest | Amazon company            |
| "Safari booking" | Wildlife tour     | Web browser               |
| "Java vacation"  | Java island       | Java programming language |
| "Cruise package" | Ship journey      | Cruise software/project   |

The system records the following user interactions:

| Query          | User Interaction                    |
|----------------|-------------------------------------|
| Amazon tour    | Viewed rainforest trekking packages |
| Safari booking | Selected wildlife resort packages   |
| Java vacation  | Viewed hotels in Bali and Java      |
| Cruise package | Viewed Mediterranean ship packages  |

However, another user searches: **"Safari download for Windows"** and clicks a browser download page.

#### Tasks:

1. Determine the intended sense of the ambiguous terms using the available context.
2. Identify the lexical and contextual cues that distinguish the possible senses.
3. Analyze why the same word may require different interpretations for different users.

4. Design an industrial-scale WSD strategy using query context, user history, embeddings, and click behaviour to improve travel recommendations.

```
print("=" * 60)
print("WORD SENSE DISAMBIGUATION")
print("TRAVEL RECOMMENDATION SYSTEM")
print("=" * 60)
```

```
queries = [
 ("Amazon tour",
 "Amazon rainforest",
 "Viewed rainforest trekking packages"),

 ("Safari booking",
 "Wildlife tour",
 "Selected wildlife resort packages"),

 ("Java vacation",
 "Java island",
 "Viewed hotels in Bali and Java"),

 ("Cruise package",
 "Ship journey",
 "Viewed Mediterranean ship packages"),

 ("Safari download for Windows",
 "Web browser",
 "Clicked a browser download page")
]
```

```
print("\nTASK 1")
```

```
for query, sense, evidence in queries:
 print("\nQuery:", query)
 print("Intended Sense:", sense)
 print("User Interaction:", evidence)
```

```
print("\nTASK 2")
print("""
Amazon + tour + rainforest -> Amazon rainforest

Safari + booking + wildlife resort -> Wildlife tour

Java + vacation + hotels -> Java island

Cruise + Mediterranean + ship -> Ship journey

Safari + download + Windows -> Web browser
""")
```

```
print("TASK 3")
print("""
The same word can have different meanings.
```

Safari can mean:

1. Wildlife safari
2. Safari web browser

For example:

Safari booking -> Wildlife tour

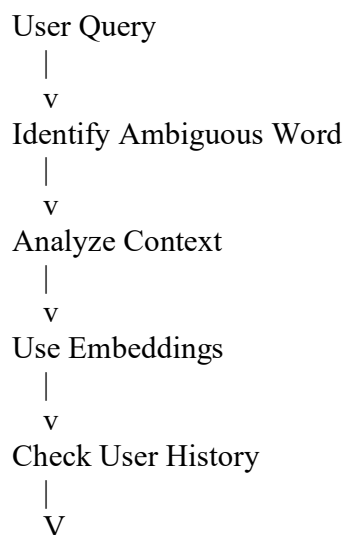
Safari download for Windows -> Web browser

Therefore, the intended meaning depends on the query,  
context and user behaviour.  
""")

```
print("TASK 4")
print("""
An industrial WSD system can use:
```

1. Query Context
2. User Search History
3. User Browsing History
4. Click Behaviour
5. Word Embeddings
6. Transformer Models such as BERT
7. Semantic Similarity
8. Recommendation Feedback

Process:



Analyze Click Behaviour

|  
v

Select Most Probable Sense

|  
v

Generate Recommendation

""")

print("=" \* 60)

print("WSD ANALYSIS COMPLETED")

print("=" \* 60)

co4.at1 q4.py - C:/Users/ADMIN/OneDrive/ドキュメント/NLP/ass python/co4.at1 q4.py (3.14.3)

File Edit Format Run Options Window Help

7. Semantic Role Labeling

8. Prepositional Phrase Analysis

9. Coreference Resolution

10. Long-Distance Dependency Handling

11. Legal Entity Recognition

12. Semantic Validation

Architecture:

Legal Document

|  
v

Sentence Segmentation

|  
v

Tokenization

|  
v

POS Tagging

|  
v

Dependency Parsing

|  
v

Voice Detection

|  
v

Semantic Role Labeling

|  
v

Legal Information Extraction

|  
v

Validation

""")

IDLE Shell 3.14.3

File Edit Shell Debug Options Window Help

TASK 2

Sentence 2 is incorrectly interpreted by the system.

Incorrect:

Buyer = Agent

Supplier = Recipient

Correct:

Buyer = Recipient

Equipment = Object

Supplier = Source/Agent

Sentence 4 is also incorrectly interpreted.

Incorrect:

Contract = Agent

Company = Object

Correct:

Contract = Object/Theme

Company = Agent

TASK 3

A grammatical subject is not always the semantic agent.

Active:

The company terminated the contract.

Company = Subject

Company = Agent

4. Syntax-Driven Semantic Analysis in Legal Document Processing

A legal NLP system analyzes sentences from contracts and assigns semantic roles based on their syntactic structures.

The system produces the following analyses:

| Legal Sentence                                             | Parse Structure      | Generated Semantic Interpretation                       |
|------------------------------------------------------------|----------------------|---------------------------------------------------------|
| "The supplier delivered the equipment to the buyer."       | Subject–Verb–Object  | Supplier = Agent, Equipment = Object, Buyer = Recipient |
| "The buyer received the equipment from the supplier."      | Subject–Verb–Object  | Buyer = Agent, Equipment = Object, Supplier = Recipient |
| "The company terminated the contract after the violation." | Subject–Verb–Object  | Company = Agent, Contract = Object                      |
| "The contract was terminated by the company."              | Passive Construction | Contract = Agent, Company = Object                      |

Tasks:

1. Analyze how syntactic structure, including active and passive constructions, influences semantic-role assignment.
2. Identify the incorrect semantic-role assignments in the given interpretations.
3. Explain how confusing grammatical subject with semantic agent can affect automated legal information extraction.
4. Recommend a syntax-driven semantic analysis architecture that can correctly handle active voice, passive voice, prepositional phrases, and long-distance dependencies in legal documents.

```
print("=" * 60)
print("SYNTAX-DRIVEN SEMANTIC ANALYSIS")
print("LEGAL DOCUMENT PROCESSING")
print("=" * 60)

sentences = [
 (
 "The supplier delivered the equipment to the buyer.",
 "Supplier = Agent, Equipment = Object, Buyer = Recipient"
),
 (
 "The buyer received the equipment from the supplier.",
 "Buyer = Recipient, Equipment = Object, Supplier = Source"
),
 (
 "The company terminated the contract after the violation.",
 "Company = Agent, Contract = Object"
),
 (
 "The contract was terminated by the company.",
 "Contract = Object/Theme, Company = Agent"
)
]
```

]

```
print("\nTASK 1")
```

```
for i, (sentence, analysis) in enumerate(sentences, 1):
 print("\nSentence", i)
 print(sentence)
 print("Semantic Roles:")
 print(analysis)
```

```
print("\nTASK 2")
```

```
print("""
Sentence 2 is incorrectly interpreted by the system.
```

Incorrect:

Buyer = Agent

Supplier = Recipient

Correct:

Buyer = Recipient

Equipment = Object

Supplier = Source/Agent

Sentence 4 is also incorrectly interpreted.

Incorrect:

Contract = Agent

Company = Object

Correct:

Contract = Object/Theme

Company = Agent

```
""")
```

```
print("TASK 3")
```

```
print("""
```

A grammatical subject is not always the semantic agent.

Active:

The company terminated the contract.

Company = Subject

Company = Agent

Passive:

The contract was terminated by the company.

Contract = Subject

Contract = Object/Theme  
Company = Agent

Confusing the grammatical subject with the semantic agent  
can cause incorrect legal information extraction.

""")

```
print("TASK 4")
```

```
print("""
```

Recommended architecture:

1. Sentence Segmentation
2. Tokenization
3. Part-of-Speech Tagging
4. Dependency Parsing
5. Constituency Parsing
6. Passive Voice Detection
7. Semantic Role Labeling
8. Prepositional Phrase Analysis
9. Coreference Resolution
10. Long-Distance Dependency Handling
11. Legal Entity Recognition
12. Semantic Validation

Architecture:

Legal Document

|

v

Sentence Segmentation

|

v

Tokenization

|

v

POS Tagging

|

v

Dependency Parsing

|

v

Voice Detection

|

v

Semantic Role Labeling

|

v

Legal Information Extraction

|

v

## Validation

```
""")
```

```
print("=" * 60)
```

```
print("LEGAL ANALYSIS COMPLETED")
```

```
print("=" * 60)
```

The screenshot shows a Python IDE with a file named 'co4 at1 q4.py'. The code defines a list of sentences and performs a legal analysis. A terminal window titled 'IDLE Shell 3.14.3' displays the output of the program, which includes a flowchart of the system architecture.

```
print("=" * 60)
print("SYNTAX-DRIVEN SEMANTIC ANALYSIS")
print("LEGAL DOCUMENT PROCESSING")
print("=" * 60)

sentences = [
 (
 "The supplier delivered the equipment to the buyer.",
 "Supplier = Agent, Equipment = Object, Buyer = Recipient"
),
 (
 "The buyer received the equipment from the supplier.",
 "Buyer = Recipient, Equipment = Object, Supplier = Source"
),
 (
 "The company terminated the contract after the violation.",
 "Company = Agent, Contract = Object"
),
 (
 "The contract was terminated by the company.",
 "Contract = Object/Theme, Company = Agent"
)
]

print("\nTASK 1")

for i, (sentence, analysis) in enumerate(sentences, 1):
 print("\nSentence", i)
 print(sentence)
 print("Semantic Roles:")
 print(analysis)

print("\nTASK 2")
print("""
Sentence 2 is incorrectly interpreted by the system.
""")
```

Flowchart of the system architecture:

```
graph TD
 A[7. Semantic Role Labeling] --> B[8. Prepositional Phrase Analysis]
 B --> C[9. Coreference Resolution]
 C --> D[10. Long-Distance Dependency Handling]
 D --> E[11. Legal Entity Recognition]
 E --> F[12. Semantic Validation]
 F --> G[Architecture:]
 G --> H[Legal Document]
 H --> I[]
 I --> J[Sentence Segmentation]
 J --> K[]
 K --> L[Tokenization]
 L --> M[]
 M --> N[POS Tagging]
 N --> O[]
 O --> P[Dependency Parsing]
 P --> Q[]
 Q --> R[Voice Detection]
 R --> S[]
 S --> T[Semantic Role Labeling]
 T --> U[]
 U --> V[Legal Information Extraction]
 V --> W[]
 W --> X[Validation]
```

## Rubrics for Calculation

| Criteria                | Weightage | Level 4 (Exemplary)                                                              | Level 3 (Proficient)                                 | Level 2 (Developing)                     | Level 1 (Beginning)                           |
|-------------------------|-----------|----------------------------------------------------------------------------------|------------------------------------------------------|------------------------------------------|-----------------------------------------------|
| Understanding of Data   | 25%       | Accurately interprets all data patterns, trends, and relationships               | Interprets data correctly with minor gaps            | Partial understanding; misses key trends | Misinterprets or fails to understand data     |
| Analysis                | 25%       | Provides deep analysis with strong logical reasoning and justification           | Provides reasonable analysis with some justification | Basic analysis with weak reasoning       | No meaningful analysis provided               |
| Application of Concepts | 20%       | Effectively applies relevant concepts/tools to interpret data accurately         | Applies concepts with minor errors                   | Limited application; requires guidance   | Unable to apply concepts to data              |
| Conclusion              | 20%       | Draws clear, meaningful conclusions with strong insights and implications        | Provides valid conclusions with moderate insights    | Conclusions are vague or weak            | No clear or valid conclusions                 |
| Clarity                 | 10%       | Presents interpretation clearly using proper structure, visuals, and terminology | Generally clear with minor issues                    | Lacks clarity or proper structure        | Poor presentation and difficult to understand |

| Q. No. / Criteria Level | Understanding of Data | Analysis | Application of Concepts | Conclusion | Clarity | Sub. Total | Total % |
|-------------------------|-----------------------|----------|-------------------------|------------|---------|------------|---------|
| 1                       | 5                     | 5        | 5                       | 5          | 5       | 25         | 93      |
| 2                       | 5                     | 5        | 5                       | 5          | 5       | 25         |         |
| 3                       | 5                     | 4        | 4                       | 4          | 4       | 21         |         |
| 4                       | 5                     | 5        | 4                       | 4          | 4       | 22         |         |

| Faculty In-charge | Course Coordinator | Program Director |
|-------------------|--------------------|------------------|
|                   |                    |                  |
| Signature: _____  | Signature: _____   | Signature: _____ |
| Date: _____       | Date: _____        | Date: _____      |